1 Weather

1.1 Temperature

AR process with a seasonal component.

Assume that temperature $T(t)$ at time t follows CARMA(p,q) dynamics for $q < p$, such that:

$$T(t) = \Lambda(t) + Y(t), \tag{1}$$

where $\Lambda(t)$ is a seasonality function (mean reversion function) and $Y(t) = b'(t)$.

- matrix A satisfies the conditions for stationarity
- volatility is periodic
- deseasonalized temperatures are stationary.

Benth and Benth (2012) - AR(3) with a seasonal component:

$$\Lambda(t) = a_0 + a_1 t + a_2 \cos\left(\frac{2\pi t}{365}\right) + a_3 \sin\left(\frac{2\pi t}{365}\right). \tag{2}$$

Cabrera [2010] - truncated Fourier series defined as:

$$\Lambda(t) = a_0 + a_1 t + \sum_{l=1}^{L}\left[a_{4l} \cos\left(\frac{2\pi(t-d_l)}{365}\right) + b_{4l} \sin\left(\frac{2\pi(t-d_l)}{365}\right)\right], \tag{3}$$

Truncated Fourier Series to capture the seasonal patterns evident in the time-varying variance of weather data:

$$f(t) = a_0 + \sum_{n=1}^{N}\left(a_n \cos\left(\frac{n\pi t}{365}\right) + b_n \sin\left(\frac{n\pi t}{365}\right)\right), \tag{4}$$

N - number of harmonics used in the approximation, a_n and b_n - the cosine and sine coefficients.

Final CARMA model fitted -

$$\Lambda_t = a_0 + a_1 t + \sum_{l=1}^{L} a_{l+1} \cos\left\{\frac{2\pi(t-d_l)}{365}\right\}, \tag{5}$$

Coefficients a_0 and a_1 respectively. Speed of the seasonal process is defined as $\frac{2\pi}{365}$. he CAR(p) process, which is the standard workhorse of modeling weather parameters, is a special case of the CARMA(p,q) process when q is 0.

1.2 Wind Dynamics

Benth (2010) - Under general transformation *wind speed* $W(t)$ at time $t \geq 0$:

$$W(t) = \begin{cases} [\lambda(\Lambda(t) + Y(t)) + 1]^{1/\lambda}, & \lambda \neq 0, \\ \exp(\Lambda(t) + Y(t)), & \lambda = 0, \end{cases} \tag{6}$$

for a constant λ and $\Lambda(t)$ a seasonal function as in the previous section. Benth (2010) - take

$$\frac{1}{\lambda} = 5$$

.

Parameter Calibration

Parameter	**Variable**	**Value**
Long-Term Mean Vector (μ)		
	speed	3.4932
	direction	145.4068
Drift Matrix (Θ)		
	Θ_1	Θ_2
speed	0.0617	0.0008
direction	0.9188	0.1953
Diffusion Matrix (Σ)		
	Σ_{11}	Σ_{12}
speed	0.9110	2.7846
direction	2.7846	1813.5325

Table 1: Multivariate Ornstein-Uhlenbeck Model Fitted to Wind speed and degree data - Weather Station 6001 (HK PolyU)

Table 2: Extreme Value model params - peaks over threshold (POT) method on Waglan Island typhoon data (1953-2006)

Parameter	**200 m**		**50 m**	
	20 m/s threshold	24 m/s threshold	20 m/s threshold	24 m/s threshold
Rate (/year), λ	1.73	1.69	1.73	1.69
Scale, σ (m/s)	6.8	9.1	6.2	8.5
Shape, k	0.041	0.031	0.0*	0.0*
20 year R.P. (m/s)	42.3	54.2	42.0	53.9
50 year R.P. (m/s)	47.5	61.5	47.7	61.6
200 year R.P. (m/s)	55.1	72.2	56.3	73.4
1000 year R.P. (m/s)	63.4	84.1	66.3	87.0
(V_{1000}/V_{50}) 2	1.78	1.87	1.93	1.99
(V_{1000}/V_{50}) 2.5	2.06	2.19	2.28	2.37

Source - Holmes et al

Peaks Over Threshold (POT) Method - This involves identifying the exceedances of wind speeds above a predefined threshold u_0. The exceedances are modeled using the Generalized Pareto Distribution (GPD), and this distribution is used to predict extreme events (e.g., the wind speeds here).

Generalized Pareto Distribution (GPD) models exceedances above each threshold u_0 as:

$$P(X_{s,t} > x) = \left(1 + \frac{x - u_0}{\sigma}\right)^{-1/k} \quad \text{for } x > u_0$$

where: - σ is the scale parameter, - k is the shape parameter, - u_0 is the threshold.

The return period R wind speed is predicted as:

$$U_R = u_0 + \frac{\sigma}{k}\left[1 - (\lambda R)^{-k}\right]$$

where: - U_R is the predicted wind speed for a return period R,

1.3 Precipitation

Similar to W Hardle (2017), a Multi-site rainfall generation model proposed by Wilks (1998) can be implemented in R. (empirical support).

- Multisite dependence with stochasticity modeled using a logit regression (Generalized Linear Model)
- Markov Chain using an optimized order based on BIC. Assuming rainfall in a single site is a Markov chain with two states, wet and dry.

Single site input with monthly as well as daily resampling of data (2010 - 2024 with varying availability per weather station.)

Calibration

Precipitation at time t in station s is modeled as - $R_{s,t} = r_{s,t}X_{s,t}$

where $X_{s,t}$ forms a Markov Chain -

$$X_{s,t} = \begin{cases} 1 & (\text{wet, } X_{min}) \\ 0 & (\text{dry, } < X_{min}) \end{cases}$$

The transition probabilities to the wet state are estimated based on the order of Markov Chain assumption. (eg: first-order assumption i.e. previous day dependence only)

$$P(X_{s,t} = 1 \mid X_{s,t-1} = 0) = p_{01,s,t} \quad \text{and} \quad P(X_{s,t} = 1 \mid X_{s,t-1} = 1) = p_{11,s,t}$$

Correlated occurrences: The neighboring locations are modeled based on empirical correlation, covariate functional - $v_{s,t}$ and $w_{s,t}$, and a pre-selected threshold probability $p_{s,crit}$. Under normal covariates, $w_{s,t} \sim N(0, \Sigma)$, with $\Sigma_{s,s'} = \text{Corr}(w_{s,t}, w_{s',t})$ and $\phi(.)$ as the CDF of standard normal distribution -

The multi-site rainfall amount conditioned on a rainy day $r_{s,t}|X_{s,t} = 1$ is assumed to follow a mixture of two exponential distributions with a time-dependent mixing parameter $\gamma_{s,t}$ and time-changing means $\beta_{1,s,t}, \beta_{2,s,t}$ (Wilks (1998))

$$r_{s,t} = r_{min} - \beta_{s,t} \log \Phi(v_{s,t}) \tag{41}$$

where:

$$\beta_{s,t} = \begin{cases} \beta_{1,s,t} & \text{if } \Phi(w_{s,t})/p_{s,crit} \leq \gamma_{s,t} \\ \beta_{2,s,t} & \text{if } \Phi(w_{s,t})/p_{s,crit} > \gamma_{s,t} \end{cases} \tag{42}$$

2 Pricing

- Non-contingent contract

$$g_t(T, T) = S(T)e^{r(T-t)}, \tag{7}$$

Under real measure P in a risk-neutral measure, yielding an a-priori random price.

- Portfolio

$$\pi_t^L(\tau_1, \tau_2) + \sum_{j=1}^{L} \gamma_j W_j(\tau_1, \tau_2), \tag{8}$$

Need weight vectors for each region while optimizing the variance-covariance matrix of the payoffs.

2.1 Hedge Effectiveness

Hedged (H), Partially hedged (HP) and unhedged (P) portfolios:

$$R_H = r_t^h(\tau_1, \tau_2) + \sum_{j=1}^{L} \gamma_j W_j(\tau_1, \tau_2), \tag{9}$$

$$R_P = r_t^h(\tau_1, \tau_2), \tag{10}$$

- cost of hedging as a percentage of the revenue / real estate invetsment $\left(\frac{\bar{R}_H - \bar{R}_P}{\bar{R}_P}\right)$
- variance reduction $\left(1 - \frac{\sigma_H^2}{\sigma_P^2}\right)$